nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Thermo Scientific Xcalibur (v3.1, v4.2)
Data analysis	Proteome Discoverer 3.1 (v3.1.0.638), Kojak (v2.1.0), MS Annika (v3.0.1), MS Amanda (v3.1.21.532), MaxLynx as part of MaxQuant (v2.6.2.0), MeroX (v2.0.1.4), pLink (v2.3.11), xiSearch (v1.7.6.7), xiFDR (v2.2.1), XlinkX (Proteome Discoverer 3.1), Xolik (v0.3), Percolator (v3.05.0), IMP-X-FDR (v1.1.1), AlphaFold2 Multimer with MMseqs (git@92deb92) and colabfold (git@7227d4c), AlphaLink2 (v2.0), xiView, the R programming language (v4.3.3), the python programming language (v3.11.9)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Raw data and result files are available via the PRIDE repository using identifiers PXD055488 and PXD055512.

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Not applicable, research did not involve human participants.
Population characteristics	Not applicable, research did not involve human participants.
Recruitment	Not applicable, research did not involve human participants.
Ethics oversight	Not applicable, research did not involve human participants.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Data was recorded in biological triplicates measured on different days using the same settings by mass spectrometry. Besides that. no calculation for sample size has been done as no biological variation is to be expected.
Data exclusions	No data was excluded from the analysis.
Replication	Reproducibility by means of crosslink numbers was estimated by calculation of standard deviations. Reproducibility by means of crosslink identifications was visualized by creating venn diagrams. Both are visualized in manuscript and supplementary material.
Randomization	Not applicable, as we do not investigate biological effects and therefore had no groups.
Blinding	As this is not a clinical study, blinding is not applicable.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	C. elegans strain: M01E11.3(jf92[M01E11.3::unc-119(+)])I; jfsi38[gfp::rmh-1;cb-unc-119(+)]II; unc-119(ed3)III;vieSi146(pAD860;pCFJ151 ppie1::SV40::vhh4GFP4::TurboID::tbb-2UTR; cb unc-119(+))IV
Authentication	Not applicable.

Mycoplasma contamination	Cells were not regularly tested for mycoplasma status.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in the study.